

Assignment 2

Deadline: 20th June 2021, 23:59:59

OVERVIEW

Design a sliding puzzle (also called [15-puzzle](#)) game website using JavaScript and your previous knowledge of HTML and CSS. Here is a [link](#) to try the game if you are unfamiliar with how it works.

TASKS

Main Tasks

[150 marks]

1. Make a classic 15-puzzle game website that allows the basic functionality for the user to start a new game and complete it. Refer to the tutorials provided at the end if you are stuck with how to start. [60]
2. Make the website responsive for all types of devices and allow all types of input viz. mouse, keyboard and touch (optional: swipe gesture) [25]
3. Add a timer and move count and maintain a leaderboard of the best times (store this data in the browser's local storage) [25]
4. Add option to choose among different sizes such as 3x3, 4x4, 5x5... (probably any MxN) and image versions of the puzzle. [25]
5. Allow multiple slides to slide at once. Make it so that if you click any square in the empty square's row or column, even ones more than one spot away from the empty square, all squares between it and the empty square slide over. [15]

Design of the website (UI/UX)

[25 marks]

You're free to decide upon the layout and design elements of your website yourself. This includes both the user experience while interacting with the website and the look and feel of the website. No need to be all showy, a minimalist design with simple animations is also enough. Bring out your creativity here. 5 marks are for sliding

animation of the puzzle blocks. [For this assignment, you're free to use CSS libraries such as Bootstrap]

Code Structure and Readability

[10 marks]

Make sure your code is Reusable, Easy to read and follow the DRY (Don't Repeat Yourself) principle. Marks will be given keeping in mind these parameters.

Additional Task

[15 marks]

Make a challenge mode of the game in which some random block/s freezes its position for a few seconds making it difficult for the player to complete the puzzle.

[10 marks + 5 marks for other innovative challenges]

RESOURCES

- ❖ <https://www.101computing.net/sliding-puzzle/>
- ❖ <https://codeguppy.com/site/tutorials/sliding-puzzle.html>

WORDS OF ADVICE

- ❖ If you are totally lost about how to do this assignment, first of all, get familiar with the basic syntax of JavaScript and then go through the above two tutorials to understand how the game is going to work. You may code along with the tutorials line by line but don't copy any code without understanding how it works.
- ❖ Create helper functions for however small operations you are going to perform such as sliding blocks, checking if the current state is a win, etc. This makes the code easy to read and make changes.
- ❖ Feel free to contact us to ask any doubts that you might face while doing the assignment.